

This repository contains the full data-processing pipeline (five scripts) used to handle technology-rich ESMs (e.g. OPERA) and prepare their outputs for sector-structured CGE models (e.g. GEM-E3) for soft-linking between the two. The workflow automates all extraction, structuring, aggregation and investment-synthesis steps.

1. [Script 1](#) extracts OPERA output into a consolidated Excel workbook.
2. [Script 2](#) processes the extracted data into a structured energy-balance dataset.
3. [Script 3](#) learns all formula references in the original IND sheet and reconstructs it entirely from raw energy data.
4. [Script 4](#) repeats the full procedure for investments: extraction, corrections, percentage allocation, multiplication, and aggregation.
5. [Script 5](#) learns the investment formulas and reconstructs the investment IND sheet from the investment dataset created in Script 4.

Script 1: `extract_ESM_energy_and_costs.ipynb`

Purpose:

Extract specific OPERA output fields from each yearly workbook and write them into `Book-EnergyBal` with `biofuels.xlsx` (energy flows) and `Inv-python.xlsx` (costs).

Input:

`Results Opera SUS NL TRANSFORM - SolarWind [year].xlsx`
Sheets used:

- “Energy balance options” (rows 6+, Columns A, B, D)
- “Cost per option” (rows 11+, Columns A, G)

Output:

Workbook: `Book-EnergyBal` with `biofuels.xlsx`
Sheets created or overwritten:

- “2025”, “2030”, “2035”, “2040”, “2045”, “2050”
Each sheet contains the cleaned OPERA energy entries (Technology, Carrier, Value).

Workbook: `Inv-python.xlsx`
Sheets created or overwritten:

- identical set of yearly sheets holding OPERA cost information.

Role in pipeline:

All later scripts depend on these yearly sheets. No later script reads OPERA (ESM) files directly.

Script 2: `structure_ESM_energy_data.ipynb`

Purpose:

Transform Script 1’s yearly OPERA extractions into sector-resolved and technology-resolved energy datasets. This includes aggregation, reshaping, and mapping of technologies to sectors.

Input:

Book-EnergyBal with biofuels.xlsx

Sheets required:

- “2025”, “2030”, ... “2050” (written by Script 1)
- “first” (technology-to-sector mapping, created manually)

Output:

Sheets written inside the same workbook:

- “Input-Technology”
- “Output-Technology”
- “Energy Input-Sector”
- “Energy Output-Sector”
- “Energy input per carrier”
- “Energy Output per carrier”
- “Tech & Carrier by Year”
- “Tech, Carrier & Sector by Year”
- “Electricity & Heat”
- “public transport”
- “private transport”

Details:

The script groups energy values by technology, carrier, sector, and year. All values ultimately originate from OPERA’s “Energy balance options” sheet (Column D), extracted in Script 1. No other source is used.

Role in pipeline:

The sheet “Tech, Carrier & Sector by Year” is the master dataset for energy-flow reconstruction in Script 3.

Script 3: learn_and_rebuild_IND_EN.ipynb

Purpose:

Rebuild the entire IND sheet (energy block) by learning the formula structure used in the original Excel model and remapping each formula reference to the structured dataset generated by Script 2.

Mechanism:

1. Parse the original IND sheet in `clean file to learn.xlsx`.
2. Extract every formula reference (sheet name + cell).
3. Translate each reference to its source in:
 - “Energy input per carrier”
 - “Tech, Carrier & Sector by Year”
 - “Electricity & Heat”
4. Identify multipliers (1000, -1000).
5. Construct a semantic mapping table: `final_formula_mapping_semantic.xlsx`.
6. Use that mapping to rebuild the full IND block from the raw structured energy data.

Output:

- `final_formula_mapping_semantic.xlsx` (mapping file)
- “IND_rebuilt” sheet written to `Book-EnergyBal` with `biofuels.xlsx`

Role in pipeline:

This script ensures the GEM-E3 linking process is fully replicable without Excel formulas, relying entirely on structured data (that can be edit by the user).

Script 4: process_investments.ipynb**Purpose:**

Extract OPERA investments per technology, remove cumulative effects, allocate investments to energy carriers using carrier shares, multiply investments by these shares, and aggregate all results by technology and sector.

Inputs:

From Script 1:

`Inv-python.xlsx` → yearly sheets (raw OPERA investment values)

From Script 2:

`Book-EnergyBal` with `biofuels_inv.xlsx` →

- “Tech, Carrier & Sector by Year”
- “Energy Carrier Percentages” (computed here)

From the mapping sheet:

- “first” (technology-to-sector mapping)

Outputs:

Sheets written to `Book-EnergyBal` with `biofuels_inv.xlsx`:

- “first” (corrected investments)
- “Sector Investments”
- “Energy Carrier Percentages”
- “[year]” (investment sheets updated)
- “Final_Investments_Multiplied”
- “Aggregated_By_Sector”

Role in pipeline:

Creates the complete, carrier-resolved investment dataset used in Script 5.

Script 5: learn_and_rebuild_IND-INV.ipynb**Purpose:**

Replicate Script 3 but for the investment block rather than the energy block.

Mechanism:

1. Parse all investment-related formulas in the original IND sheet.
2. Extract all references to:
 - “Aggregated_By_Sector”
 - “Final_Investments_Multiplied”
3. Build a semantic mapping file:
`final_formula_mapping_semantic-INV.xlsx`
4. Recalculate every IND investment cell numerically using the datasets produced by Script 4.

Outputs:

- `final_formula_mapping_semantic-INV.xlsx`
- “IND_rebuilt” sheet written into Book-EnergyBal with `biofuels_inv.xlsx`

Role in pipeline:

Provides a fully formula-free reconstruction of the entire investment block, aligned with the GEM-E3 model inputs.

File Overview

Extracted Data:

- Book-EnergyBal with `biofuels.xlsx` (energy extraction and processing)
- Inv-python.xlsx (raw investment extraction)

Structured Data:

- Book-EnergyBal with `biofuels_inv.xlsx` (all carrier-resolved investment data)

Mapping Files:

- `final_formula_mapping_semantic.xlsx`
- `final_formula_mapping_semantic-INV.xlsx`

Original Learning Files:

- `clean file to learn.xlsx` (for energy)
- `INV clean file to learn.xlsx` (for investments)